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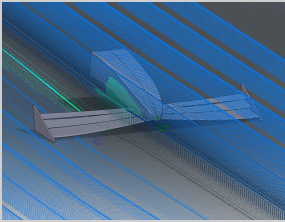
Mechanical & Aerospace Engineering – Cornell University, Expected May 2028

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I build and test hardware myself – from CAD and CFD through fabrication, integration, and field or flight test – rather than stopping at a design review. That ownership runs from a classified defense enclosure now planned for fielded deployment, to UAVs I've personally built, wired, and flown, to a self-taught CFD study measuring real aerodynamic loads on a wing I designed from scratch.

Full project write-ups, CAD files, and build photos/videos available at the site linked above.

SHEET 01 PROJECTS

Classified Hardware Enclosure ICR, INC. Owned a classified defense hardware enclosure from concept to fielded systems — independently leading CAD, CFD/FEA analysis, and production drawings for a multilayer protective enclosure balancing thermal, structural, and sealing requirements. Now planned for integration across multiple fielded defense systems.	F1 Front Wing Design & CFD PERSONAL PROJECT  Self-initiated a deep dive into SolidWorks Premium's Flow Simulation tool, designing a multi-element F1-style front wing from scratch with no class or team behind it. Ran an external flow analysis at 100 mph and measured 347.5 N (≈78 lbf) of downforce by integrating pressure loads across the wing.
 UAV Systems LOCKHEED MARTIN × NORTHFIELD HIGH SCHOOL / PERSONAL Built and flew UAVs hands-on, not just designed them: as a Northfield High School (Denver, CO) student, worked in tandem with Lockheed Martin engineers to build a LIDAR-mapping hexacopter, running onboard Raspberry Pi/LIDAR mapping software myself. Then cut, wired, and flight-tested a scratch-built foamboard fixed-wing aircraft through several prototypes until it flew clean.	Pulse Power CORNELL MAKEATHON 2026 Designed, built, and demoed a wearable sleep-stage-detection device in a single 24-hour hackathon, using heart rate and skin temperature sensing to wake the user at the end of a REM cycle. Built with teammates Sani Mohottige and Liam Alcalay.
Makerspace Builds CORNELL MAKER CLUB Ongoing personal robotics, mechatronics, and UAV-component projects in the Maker Club lab — motor integration, custom gearboxes, and rapid-prototyped enclosures using CNC, electronics, and 3D printing.	Vibrating Grape Filtration CORNELL — CLASS PROJECT Designed and built the motor-driven vibration linkage for a filtration tray that separates spotted lanternfly contamination from grape juice during harvest, sourcing components from McMaster-Carr. Built with Flynn Basehart, Kathryn Moskal, Emma Harshberger, and Gavin Sneyd.

Cornell Maker Club Executive Board (Jan 2026–Present) – overseeing makerspace lab operations and training members on machinery and safety protocols – and **Red Apron certified** (mill & lathe), reflecting ongoing hands-on systems work and technical leadership beyond individual projects.